MIL-F-47249(MI)
26 July 1974
SUPERSEDING
MIS-13981
2 April 1969

MILITARY SPECIFICATION

FILM, POLYAMIDE

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of polyamide film.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL- P-14232	Parts, Equipment, and Tools for Army Materiel, Packaging and Packing of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC 8135

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials

ASTM D 645	Thickness of Paper and Paper Products
ASTM D 882	Tensile Properties of Thin Plastic Sheeting
ASTM D 1004	Tear Resistance of Plastic Film and Sheeting
ASTM D 1204	Measuring Changes in Linear Dimensions of Nonrigid Thermoplastic Sheeting or Film
ASTM D 1505	Density of Plastics by the Density-Gradient Technique
ASTM D 1790	Brittleness Temperature of Plastic Film by Impact
ASTM D 1922	Propagation Tear Resistance of Plastic Film and Thin Sheeting
ASTM E 96	Water Vapor Transmission of Materials in Sheet Form

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

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Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample shall meet all the requirements of this specification.

3.2 Material.

3.2.1 Composition. The material shall be a nylon-6-type thermoplastic film made from a polymer of -caprolactam, treated both sides for b ndability.

3.2.2 Thickness. Film thickness shall be 0.001 plus or minus 0.0002 inch.

3.3 Performance characteristics. The finished film shall conform to the requirements specified in Table I.

Table I

Performance characteristics	
Property	Requirements
Specific gravity	1.13 ± 0.02
Tensile strength MD XD	9,000 psi (minimum) 10,000 psi (minimum)
Tear strength, initial (Graves) MD XD	300 gm/mil (minimum) 300 gm/mil (minimum)
Tear strength, propagated (Elemendorf) MD XD	25 gm (minimum) 25 gm (minimum)

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Table I continued

Dimensional change (Shrinkage) MD XD	3 percent (maximum) 3 percent (maximum)
Cold Brittleness	lower than minus 53.8°C (minus 65°F)
WVTR	25 gm/100 sq in/24 hrs (maximum)

* MD = Machine direction

XD = Transverse direction

3.4 Workmanship. The workmanship shall be such as to ensure a product which is uniform and in accordance with this specification. The film shall be clean, well finished and free from dirt, oil, foreign matter, rough or sharp edges, scratches, scuffs, cracks, creases, blisters, bubbles, pimples, pits, tears, cuts and holes, which might impair its usefulness for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. Unless otherwise specified in the contractual document (see 6.2), a sample consisting of 4 square feet manufactured under the same conditions as those proposed for subsequent production, shall be subjected to preproduction inspection. Accomplishment of preproduction inspection shall be as specified herein.

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Subsequent units will not be considered for acceptance until Government approval of the preproduction sample has been obtained. Units subjected to preproduction inspection shall have successfully passed acceptance inspection. Testing of the preproduction sample to determine compliance with the characteristics listed in Table II shall be conducted in accordance with the corresponding test paragraphs.

4.3 Quality conformance inspection.

4.3.1 Sampling for acceptance inspection. Sampling for acceptance inspection shall be conducted on a random sample selected from each inspection lot or less in accordance with MIL-STD-105 at an acceptable quality level (AQL) of 4.0 percent defective.

Table II

Preproduction inspection		
Characteristic	Requirement Source	Test Paragraph
Specific gravity	3.3	4.5.3.2
Tensile strength	3.3	4.5.3.3
Tear strength, initial	3.3	4.5.3.4
Tear strength, propagated	3.3	4.5.3.5
Dimensional change	3.3	4.5.3.6
Cold brittleness	3.3	4.5.3.7
WVTR	3.3	4.5.3.8

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4.3.2 Lot formation. A lot shall consist of all the material manufactured in one continuous operation by the same process by the same manufacturer in accordance with this specification and submitted for inspection at one time.

4.3.3 Acceptance inspection. Acceptance inspection of a sample as specified in 4.3.1 to determine compliance with the characteristics specified in Table III shall be conducted in accordance with the corresponding test and inspection paragraphs.

Table III

Acceptance inspection		
Characteristic	Requirement Source	Test Paragraph
Composition	3.2.1	4.5.3.1
Thickness	3.2.2	4.5.3.9
Workmanship	3.4	4.5.5
Packaging and packing	5.1	4.5.4
Marking	5.2	4.5.4

4.4 Inspection equipment. The inspection equipment for conducting examination and tests shall be as specified in the applicable test methods and procedures paragraphs.

4.5 Test methods and procedures.

4.5.1 Test condition. Unless otherwise specified herein, the following conditions shall be used as a basis to establish performance requirements:

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a. Temperature, room ambient (16 to 32 degrees Celsius (C)) (60 to 90 degrees Fahrenheit (F)).

b. Altitude, facility ground.

c. Humidity, facility ambient up to 95 percent relative humidity.

4.5.2 Test sequence. Test sequence within each classification of inspection (preproduction or acceptance) shall be at the option of the supplier.

4.5.3 Tests.

4.5.3.1 Composition. The composition shall be certified by the supplier to be in conformance to 3.2.1.

4.5.3.2 Specific gravity. The specific gravity shall be determined in accordance with ASTM D 1505 and shall conform to 3.3.

4.5.3.3 Tensile strength. Tensile strength shall be determined in accordance with ASTM D 882 and shall conform to 3.3.

4.5.3.4 Tear strength, initial. The initial tear resistance shall be determined in accordance with ASTM D 1004 and shall conform to 3.3.

4.5.3.5 Tear strength, propagated. The propagated tear resistance shall be determined in accordance with ASTM D 1922 and shall conform to 3.3.

4.5.3.6 Dimensional change. The percent dimensional change shall be determined in accordance with ASTM D 1204 after a 30 minute exposure in an air circulating oven at 149 degrees C (300 degrees F) and shall conform to 3.3.

4.5.3.7 Cold brittleness. The cold brittleness temperature shall be determined in accordance with ASTM D 1790 and shall conform to 3.3.

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4.5.3.8 Water vapor transmission rate. The WVTR shall be determined in accordance with ASTM E 96 and shall conform to 3.3.

4.5.3.9 Thickness. Thickness shall be determined in accordance with ASTM D 645 and shall conform to 3.2.2.

4.5.4 Packaging, packing and marking. The inspector shall ascertain that the packaging, packing and marking of the material conforms to this specification.

4.5.5 Workmanship. The film material shall be visually examined to assure conformance to the workmanship requirements specified in 3.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. The material shall be furnished in rolls, as specified by the procuring agency, from the range of length, width, core diameter, allowable tolerances, and number of splices normally available from the supplier (see 6.2). The polyamide film material shall be packaged and packed in accordance with specified level of MIL-P-14232 and in such a manner as to preclude damage or contamination during handling, transportation, or storage.

5.2 Marking. Marking shall be in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

a. Manufacturer's name and address.

b. Manufacturer's designation.

c. Lot or batch number.

d. Date of manufacture.

e. Number of this specification.

6. NOTES

6.1 Intended use. The polyamide film material in accordance with this specification is intended for use in

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the construction of a moisture barrier fly-through diaphragm in a missile container to provide protection against corrosion and deterioration prior to missile launch.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required (see 3.1) and, if so, pertinent details (see 4.2).

c. Quantity and size (see 5.1).

d. Requirement for certification of composition (see 4.5.3.1).

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-13981, dated 2 April 1969.

Custodian:
Army-MI

Preparing Activity:
Army-MI
Project No. 8135-A435

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CONTRACT NUMBER

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☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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